Data Manipulation and Analysis -- Elective

Course Description

This course introduces students to the emerging field of Data Science. Instructional units cover the standard practices for effective data manipulation, analysis and interpretation as well as necessary concepts in the three disciplines involved (mathematics, statistics and computing.) Numerous examples of typical problems encountered in Big Data are provided. The emphasis on this course is in the application of the concepts rather than the theory. In the second semester, students will work in teams on large projects in which they will use programming to analyze large datasets and create predictive models. The students will summarize their findings for each project in a written report and will also present them orally.

Course Objectives

- Create models driven by large amounts of data
- Interpret graphs and tables, and make sense of data by visualizing it
- Create interactive presentations for visualizing data
- Automate routine data manipulation tasks, such as formatting, collection and storage
- Describe basic algorithms for clustering, regression and classification of data.
- Explain typical trade-offs that data scientists must make when analyzing data.
- Demonstrate effective communication skills, through team working, oral presentations and good written communication.

Assessing Performance

In the first semester, formative assessment includes worksheets, several practice activities for each lesson, and unit quizzes, and summative assessment includes a small programming project at the end of each four-week unit. In the second semester, projects are assessed on accomplishment, originality, code sophistication, presentation and team work. Milestones are set for each project, so that teams receive feedback on progress before the final evaluation of each project.

Course Essentials

Equipment	Cost/Unit
Classroom set of computers	$0 if you already have some, $500-600 per computer if you need to purchase

First Semester

Unit 1: Overview of Data Science	Small data and big data. Sources of data. The curse of dimensionality. Mean and variance. Overfitting and underfitting. The Variance-Bias trade-off.
Unit 2: Review of Python and Scipy	Syntax. Semantics. Conditions. Loops. Lists. The Scipy suite of packages.
Unit 3: Data visualization	Static graphs and charts. Interactive graphs and live feeds. Visualization for exploration vs. visualization for presentation.
Unit 4: Data collection and organization	Tables, matrices and databases. Text files and the CSV format. Web scraping. JSON and XML Web APIs. Cleaning and summarizing data.

Second Semester

Unit 5: Exploratory data analysis	Projects will use either Descriptive statistics or Inferential statistics (Hypothesis testing) to explore large datasets of interest to the students.
Unit 6: Regression	Students will analyze datasets to create linear models that describe relationships between features of interest.
Unit 7: Clustering	Students will use clustering techniques, such as k-means, to divide the data into meaningful subsets. Students will explore the effect of changing the number of clusters and will try to find an optimal subdivision.
Unit 8: Classification	Students will run training sets of data using decision trees, naive Bayes or, in some cases, neural networks that learn a predictive model fitting the data.

DATA MANIPULATION AND ANALYSIS

1. Materials
   Internet access and 1-to-1 computer use daily.
   Chromebooks, Windows PCs, and Mac computers are acceptable for this course.
2. Required software, networking access, and access to LSU resources:
   - Students and teachers must have reliable access to the Louisiana Training Platform (LTP), louisianatrainingplatform.com, and district networks and student devices must allow access to the platform and its Python coding activities.
   - Course lessons, activities, Python coding practice, assignments, quizzes, and projects are delivered through LTP.
   - Students must have access to Google Sheets or a compatible spreadsheet application, such as Microsoft Excel, for spreadsheet-based assignments that may be downloaded, completed, and submitted through LTP.
   - A current web browser capable of running the course content in LTP is required.
3. Requirements for IBC Alignment: **CIW Data Analyst**
   For schools who are pursuing the aligned IBC for this course
   a. Schools must have the appropriate testing software installed on the testing computers. For this credential the testing software is through CIW. More information can be found here: https://ciwcertified.com/exam-requirements/
   b. Some industry based credential providers also provide practice tests softwares such a GMetrix which may be installed. Please check with individual schools if they plan to use a practice software.
   c. Schools must also typically register as an approved testing center prior to being allowed to proctor the exams.
   d. Teachers must have earned the IBC prior to students sitting for the exam. Once they have passed the credential, they need to be set up as a proctor in the CIW system which must be done by the CIW team. Please contact your regional CIW representative.
   e. Teachers should report to the LSU Point of Contact their overall pass rates (i.e. 15/20 students passed)

4. Required teacher collaborations
   Teachers will communicate with LSU instructors via email. Teachers may be asked to share student work, course implementation feedback, or other instructional information with their designated LSU Pathways Point-of-Contact.

5. Required administration of course content, pre/post test, and research instruments
   Required course materials, assessments, activities, and instructional resources will be distributed through the Louisiana Training Platform or other LSU-approved systems identified by the course instructor.

6. Course Work
   Teachers must present the course material in sequence or as approved by collaboration with the LSU Pathway Point-of-Contact. Teachers are expected to deliver a minimum of 80% of the course material for course fidelity and for the opportunity for IBC attainment (where applicable).

7. Other

As this is a hands-on, applied data analysis course, we strongly suggest that each section be limited to a *maximum* of 20 students. The course includes data-analysis activities, spreadsheet work, Python coding, and applied projects that require teacher feedback and support. Teachers should have adequate time to review student work, provide feedback, and interact with individual students.